

SAGNIK ROY

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Education

Indian Statistical Institute, Kolkata

Bachelor of Statistics (2nd Year)

Kolkata, IN

July 2024 – Present

- Core Interests: Data Science & Applied Statistics
- Current Aggregate: 89.67%

Delhi Public School, Ruby Park

Standard 12 (CBSE) – High School Equivalent

Kolkata, IN

May 2024

- Grade: 97.4%

South Point High School

Standard 10 (CBSE) – Secondary School Equivalent

Kolkata, IN

May 2022

- Grade: 98.6%

Interests

Statistical Ecology, Bayesian Statistics, Machine Learning, Interpretability of AI Models, Actuarial Modeling

Projects

Myeloid KLF4 in Lung Fibrosis — ISI, Kolkata

January 2026 – Present

- Analyzing single-cell RNA-seq data to study the role of KLF4 in myeloid cells and its association with fibroblast activation in lung fibrosis. Performing data integration, normalization, and cell-type-specific expression analysis using Seurat, focusing on KLF4 and collagen genes (COL1A1, COL3A1) to investigate myeloid-fibroblast interactions and extracellular matrix remodeling.

Lag-Aware Statistical Modelling of Tiger Mortality and Growth — Senckenberg Biodiversity and Climate Research Centre

December 2025 – Present

- Developing statistical models to study lagged relationships between tiger mortality and population growth in India.
- Investigating deviations from natural population regulation to identify potential anthropogenic influences, including poaching and habitat disturbance.

Introduction to Bayesian Data Analysis — ISI, Kolkata

May 2025 – July 2025

- Undertaking a reading project on the fundamentals of Bayesian Inference, Construction of priors, Model Selection, Robust Bayesian Outlier Detection and Computational Methods (MCMC).

CDI & MMI Prediction of Earthquakes — Impact-Metrics (IIT-BHU)

Dec 2024 – Jan 2025

- Engineered a Random Forest model to predict the CDI (Community Density Intensity) and MMI (Modified Mercalli Intensity) of earthquakes. The prediction model ensured that CDI and MMI can be calculated remotely without visiting the site and provide a better estimate of the damage caused.
- Achieved an accuracy score of 95% and 90% in the test dataset for CDI and MMI respectively.
- Constructed an ML model to evaluate and flag alert systems that underestimate an earthquake's intensity with an accuracy of 70%.
- **Achieved 2nd place in the competition.**

Stock Prediction from Tweets — Technalytics (IIT-BHU)

Jan 2025 – Feb 2025

- Applied Sentiment Analysis and Named Entity Recognition to analyze tweets and predict stock price outcomes.
- Classified keywords (companies, personalities, etc.) as high-risk, low-risk, or stable.
- Used a Logistic Regression model to fit a prediction model to predict stock prices from tweets and achieved 70% accuracy.
- **Secured 2nd place in the competition.**

Achievements

- **Jagadis Bose National Science Talent Search (JBNSTS), 2023:** Received "Encouragement Award" from the Govt. of West Bengal, given to top science students to motivate research interest.
- **ISI Entrance 2024:** All India Rank 71.
- **American Mathematics Competition (AMC 12):** School Rank 2 (12A & 12B), Country Rank 71 (12B).
- Qualified Indian Olympiad Qualifier in Math (IOQM) in 2021 & 2023.
- Qualified National Standard Examination in Chemistry (NSEC) in 2023.

Relevant Coursework(Taken & Present)

- **Probability Theory I, II & III:** Random Variables, Discrete & Continuous Distributions, Expectation, Variance, CLT, Convergence, Lebesgue Integral.
- **Stochastic Processes:** Discrete Markov chains with countable state space, Introduction to MCMC, perfect sampling
- **Statistical Methods I, II, III & IV:** Linear & Logistic Regression, Estimation (MLE, Moments), Hypothesis Testing, EM Algorithm, Steepest Gradient Descent, Multivariate Analysis, Bootstrapping Techniques, Time Series Analysis .
- **Analysis I, II & III:** Real Analysis (Sequences, Series, Integration), Multivariable Calculus, Optimization, Fourier Series.
- **Vectors & Matrices I & II:** Vector Spaces, Linear Maps, Matrix Properties & Decompositions (LU), Eigenanalysis, Quadratic Forms.
- **Programming & Data Structures:** C & Python fundamentals; Arrays, Pointers, Stacks, Queues, Linked Lists.
- **Molecular Biology:** Replication, transcription, translation, cell division (mitosis, meiosis), Definition of gene and genetic code; relationship between them Mendel's Law of genetics and application in human population

Skills

- **Languages:** English (Bilingual), Bengali (Native), Hindi (Bilingual).
- **Programming:** C, Python, R.
- **Libraries & Tools:** NumPy, Pandas, Scikit-learn, TensorFlow, Matplotlib, Seaborn, dplyr (R).